

INTRODUCTION VINCANNE ADAMS

The shift from *international health development* to *global health* in the sixty-year-old postcolonial infrastructure of transnational health aid is not a simple case of new bottles for old wine. Emergent trends attending to the desires for global health reveal complex transformations in the practices of audit, funding, and intervention. One of the most important features of this shift has been the growing push for and reliance upon specific kinds of quantitative metrics that make use of evidence-based statistical measures, experimental research platforms, and cost-effectiveness rubrics for even the most intractable health problems and most promising interventions. Anthropologists have been writing in interesting ways about these shifts, and we sense a need for both applause and caution in the embrace of these rubrics for doing health work in global ways. This book offers a series of ethnographic explorations of these trends, focusing on the on-the-ground realities of what we might call *metrics work* while also investigating the expectations and accomplishments that the *metrics* create, for better or worse, in global health today.

Global Health: Rupture and Persistence

The phrase *global health* is used to name a variety of familiar and unfamiliar activities, most of which are in perpetual motion. Chapter 1 offers a more complete overview of the central argument of the book, mapping the tra-

jectories metrics have taken in the conceptualization of global health and serving as a longer orienting chapter for the collection. Here it is useful to note that in this volume we see global health as *both* a response to exigencies of finance, data, and outcomes that have defined the postwar world of international health *and* a utopian proposition to escape, if not transcend, the problems and perceived lack of success of these arrangements and exigencies. The links among economy, sovereignty, and the politics of knowledge that have shaped the use of metrics over the past decades in this field suggest that global health strives toward forms of knowledge that are distinguished from those found in an era of postwar, postcolonial international health. It is worth noting at the outset that this claim invites scrutiny from a number of positions and empirical grounds.

Recall, for instance, that postwar international health was, in its own way, a response to what came before it: colonial health and the demands of empire that such health was attendant upon. How global health might be seen as rupture rather than continuity and the specific resemblances it has to earlier forms of health aid in *both* the colonial and postcolonial eras is perhaps most visible not just in the efforts to count things but in how counting things has worked in relation to other political and economic institutions.

For instance, global health is not the first to think “globally.” Colonial tropical medicine configured the global in its own ways. Similarly postwar international health efforts imagined a worldly set of impacts and interventions. There are differences, however, in how these eras imagined their geographic reach and how they undertook to count health, especially in relation to economics. These differences help us distinguish what is new and what is old in today’s global health metrics.

Colonial tropical disease maps of the spread of yellow fever, dengue, tuberculosis, and smallpox, and the mapping sciences deployed to measure these, were crafted under the assumption that the reach of empire went beyond sovereign borders (Anderson 2006; Tilley 2011). Colonial health efforts not only imagined geographic continuities across continents and oceans but also across racial, ethnic, and national territories, and intervention and mapping projects counted in ways that authorized and ensured the expansion and stability of these empires (Birn et al. 2009; Hacking 1990; Packard 1989, 1997a; Scott 1999).

In contrast, postwar postcolonial health development efforts were born in part as a critique of these regimes, and, to some extent, they reformulated

the possibilities of intervention in and around the birth of the postcolonial nation-state as a *response* to empire (Bose 1997; Nandy 1989; Stepan 2011). Multilateral and bilateral forms of health aid that came into being after World War II continued to count things like the spread of disease, the loss of productivity to disease, population health, and public health, but, in the aftermath of empires, they increasingly had to do so in relation to the unit of the nation state (through national ministries of health that were tasked with producing national health statistics; Greene 2008; Rees 2014).

In fact, organizations like the World Health Organization (WHO) often struggled to forge a path that transcended national interests (Chorev 2012), even while newly independent nations strove to assert claims to independence in part through health and health care (Osseo-Asare 2014). So while postwar institutions that grew in the decades after decolonization believed in a transcendent form of policymaking in which *international* bodies would set aside their national interests and create a world in which a flourishing economy could produce an even playing field, where distributions of the world's health (and health care resources) would be equitable across nations, and where the possibilities for another world war would be reduced (Finnemore 1997; Staples 2006), this process was often fraught with political divides between North and South, poor and wealthy, socialist and capitalist nations (Chorev 2012). Their worldwide goals, in part, resurrected those of earlier pan-state health agencies (such as those at the International Sanitary Convention in Paris in 1851 or at agencies like the Pan American Health Organization, founded in 1902) in their desires for international health equity in relation to both controlling the spread of disease and/or eradication of them (Stepan 2011). Even when these postwar institutions were organized around the principle of "one nation, one vote" (as the WHO was), Chorev (2012) reminds us that national concerns remained more visible than ever and, in fact, had to be managed carefully. Global health, as we will see, attempts, or imagines, something new.

Continuity and rupture over this early transition from colonial to postcolonial public health can also be traced in relation to how these eras and institutions envisioned the relationships between health and economics. Colonial health programs gave birth to statistics practices that configured population health in relation to colonial economies. Even colonial-era missionaries and humanitarian organizations were, in their own ways, focused in some measure on counting things, even if it was "souls saved" as much as "laboring bodies" (Brown 1979; Packard 1989; Rose Hunt 1999; Stepan

2011; Vaughan 2001). Health was also intertwined with practices that exceeded economics, such as when it enabled the colonized, or the postcolonial, to serve as a subject of science (Anderson 2006; Prakash 1999; Tilley 2011) and when it was authorized by and configured as humanitarian service (Birn 2014; Brown 1976, 1979; Chandler 2001). Counting health here often meant calibrating the impact to extractive industries due to things like tuberculosis, malaria, hookworm, even infertility (Brown 1979; Packard 1989, 2011; Rose Hunt 1999).

In the postwar postcolonial era efforts to control or eradicate disease and distribute health resources shifted the priorities between health and economics. If the health of natives was important to colonial administrators because of the need for labor in colonial industries even when carried out under the guise of benevolence, then in the postcolonial postwar era the assumption early on was that the health aspired to through development programs was of value in its own right (and that, indeed, impoverishment from colonial labor was often to blame for poor health). *Still, in postwar international health work it was often assumed that health could be achieved only through simultaneous economic development, that with economic development health improvements would follow seamlessly* (Bose 1997; Farley 2003; Packard 1997a, 1997b). In the cold war environment, the focus on economic development over other routes to health sometimes masked and at other times exacerbated larger debates over the benefits of socialism versus capitalism in provisioning “health for all.”

The blurred lines of cause and effect between health and economic development (that is, over whether to conceptualize health or economics as the top priority) at places like the Rockefeller Foundation (long before the postwar era) were inherited by many international health institutions that aimed to accomplish what colonial medicine failed to accomplish — namely to eradicate diseases and bring health to the natives (Birn 2014; Farley 2006; Stepan 2011). But even where the cause-and-effect relations between economic and health goals were reversed, the grand arc of health development (from colonial times through the postwar era) demonstrates a striking persistence of thinking in more foundational ways about health in terms of human productivity in economic terms.

One might argue that in the postwar era the shifting links between health and economic development priorities became more troublesome than ever, forming a source of extraordinary tension and debate within multilateral institutions like the WHO and in the various bilateral aid agen-

cies. One could even talk about shifts in postwar international health policy solely in relation to these battles between economics and health. The political economic assumptions of the Alma Ata Summit and the primary health care movement are often set against the policies of structural adjustment programs as both political economic agendas forged paths through the half-century of cold war diplomacy (Chorev 2012). Still, at the end of the day, these debates have frequently dissolved into just that: combat over one kind of political-economic policy versus another in relation to health policy. Seldom have international health institutions been able to sustain the more bold (some would say naïve) mission of promising health as a human right, regardless of cost or any particular political-economic theory.

In sum, how postwar world health institutions have gone about measuring things in relation to health is in some part continuous with efforts seen during the colonial era and in some part quite different. Efforts to eradicate diseases in the colonial epoch were largely replicated in early postwar institutions (Farley 2003; Packard 1997a, 1997b, 2011; Stepan 2011) and reproduced many of the debates over the links between economics and health. At the same time, concerns over health under empire were supplanted in the postcolonial era by concerns over health conceptualized in and through the nation-state. Similarly some of the links between economics and health configured under colonialism became reconstituted in the postwar period in policies that linked health development to economic development, even while new formulations of “health as a right” were augmented. All of these efforts and transitions were accompanied by ways of counting that fit the administrative and worldly aspirations of these times.

In contrast, *global health* is often construed as having the capacity (and focus) to transcend both the nation-state and the endless deliberation over which politics best serve to advance health in relation to economic development. Not surprisingly, it also imagines having the ability to count things differently. This is explored more fully in chapter 1. Briefly, however, the shift away from national-level discourse is in part because of the crisis resulting from the decline of fiscal support from national entities (Birn 2014; McGoey et al. 2011) and from the growing sentiment that national entities often *get in the way* of effective health delivery rather than promoting it. This latter sentiment is nested inside more critical perspectives from within global health institutions today that conceptualize both disease and interventions in ways that are believed to transcend national boundaries because they are not “political” (Rees 2014). This has not, however, entailed

transcending thinking of health in and through economics or economical ways of counting, as we will see.

Global health is thus imagined as a rubric for doing something that has not been done before: to imagine health in a truly global way, to do health work in ways that are not dependent upon the nation-state and, as we will see, that put the old stale debates over economics and politics to rest. Global health planners envision a world of diseases constantly in motion, always on the move to new places without regard to nation-state capacities and limitations. Global health professionals also envision a world in which interventions can be mapped out as problems of scale and measurement (or specifically as counting practices) rather than as problems of custom, culture, or national political will. The dominant discourses in global health today also imagine a world in which neoliberal economic strategies will seamlessly work toward the achievement of health as a right of, and therefore achievable for, all. What enables all this to happen today, of course, is metrics.

Today key institutions of global health such as the Global Fund, the Bill and Melinda Gates Foundation, and the Institute on Health Metrics and Evaluation all envision using a form of global knowledge that is based on universals (biology, disease, vaccines, etc.), in which multiplicity is visible only in and through global (that is, universal) forms of data production that get lumped together as “metrics.” Metrics are technologies of counting, but specifically technologies of counting that form global knowledge. Metrics used today are imagined to offer uniform and standardized conversations about how best to intervene, how best to conceptualize health and disease, how best to both count and be accountable, and how best to pay for it all.

One of the goals of this volume is to explore how global health metrics pose a problem of knowledge in relation to our understanding of not only the past but also the future or what might be called *global health efficacy*. Specifically, how do counting efforts seen in the turn to randomized controlled trials (RCTs), and its related forms of evidence-based practice, become the new arbiters of what counts in relation to health outcomes? How does the demand for counting in these ways impact what health workers do? How do efforts to produce RCT-like statistical validity get undermined by more basic problems of counting births, deaths, disease, and morbidity at all? How do these challenges pose problems for nation-states, for local, regional, or national governments?

The chapters here provide evidence that global health metrics enable

and encourage efforts to produce health in ways that sometimes work well and sometimes do not. We witness celebration of *the new* coupled with frequent exasperation over the recurring sense that *we have been here before*. Indeed global knowledge is fraught with many of the same problems of inclusion and exclusion (of evidence, people, problems) that were seen in the era of colonialism and in the era of postwar development aid. Still the metrics of global health authorize new demands for all of us involved in this work, and they produce effects far beyond their specific numbers. These demands and effects can be read in relation to debates about sovereignty.

Metrics and Sovereignty

What do health workers do when they cannot produce the kinds of numbers that give statistically robust results? What do they do when deaths mess up the numbers needed to claim health program success? What light do these activities and failures shed on the ways sovereignty and global health metrics work together today? The push for better ways of counting and better indices of accountability in what has been called evidence-based medicine is not new (as chapter 1 will discuss). However, the call for such tactics in global health today seems to often rely on a kind of forgetfulness, or perhaps a studied ignorance, of the past and the many efforts to critique these tactics in the Global North.

To the extent that expectations for metrics are promoted today at the largest institutions for health aid, and to the extent that funding for global health work is increasingly tied to these metrics, we explore in this volume the way the effort to “do metrics” contributes to forms of sovereignty that may ultimately challenge older forms of political sovereignty that emerged in the postwar era. Thus if we think of sovereignty in relation to the mid-seventeenth-century notion of Westphalian territorial governance, then what do we call it when a sovereign nation’s ability to govern becomes increasingly dominated not just by the regimes of development aid that reiterated its national borders (indeed its right to national forms of knowledge) but by those that aim to transcend these borders? This volume begins to answer this question or, at least, opens that conversation.

In his 1999 book, *Seeing Like the State*, James Scott noted that the degree to which modern governments succeed or fail depends in part on their ability to create systems of quantification that render complex social phenomena comparable and countable. Postcolonial international health

could be seen as having given rise to quantification practices that leveraged the use of international resources for state building. Metrics-driven political economic arrangements today, however, may promote infringements on state sovereignty in and through global health regimes. The degree to which these forms of counting ironically reproduce older forms seen under colonialism is a concern here. The wide range of phenomena that are pushed inside and outside of visibility in the wake of the demand for metrics, and how these acts of recognition, exclusion, and inclusion draw our attention to problems of not only epistemology and economics but also politics and governance are all important. This volume provides studies in the trials and tribulations of a multidisciplinary effort to *do global health through metrics* but also opens the conversation about what else is at stake in these efforts in relation to an aspirational postcolonial, but also post international health—that is, post–nation-state—form of global health sovereignty.

The use of the term *metrics* in this volume is itself somewhat aspirational. Our papers suggest that the ideal form of metrics used in accounting today in global health programs is in fact often more hoped-for than achieved. Indeed the counting practices of today are as troubled by imperfection as were the efforts in previous times. This does not, however, get in the way of efforts to produce them, nor does it impede efforts to rely on their empirical products as if they were indelibly factual. One of the attractions of metrics is their ability to hold status as apolitical or politically neutral forms of evidence. Of course, no history of metrics would suggest that counting exercises are not deeply entangled with politics. Still, the assumption still seems to be in many global health circles that numbers will offer unbiased, apolitical truths about health outcomes or health conditions. Our focus here, on the specific kinds of metrics that are espoused at today's global health institutions and in global health efforts all over the world, is to trace these productivities and to specify how they conceal, reveal, and generate both success and failure when it comes to problems of both politics and health.

The authors here note that not only are reliable numbers both frequently impossible to get and frequently misleading in the world of global health but also that other forms of evidence may be *more* reliable than those that can be rendered through metrics. Getting the empirical facts needed for metrics may entail a kind of violence to the empirical truths they aim to produce. Other kinds of evidence may be more reliable than numbers. In this sense, quantification strategies and the metrics we rely on to *avoid* poli-

tics often do not avoid politics at all; they become a form of politics in their own right, augmenting the political stakes and political underpinnings of health projects in a manner that is frequently invisible to those who believe in these exercises in calculation and counting. In this sense the notion that metrics are not (and never have been) politically neutral is not only worth repeating, but also exploring further.

Most of us (and certainly not only anthropologists) know that numbers are never intrinsically capable of proving anything; they must be made to speak in very specific ways about what they claim to represent (Crump 1990). In fact we might see number crunching and metrics work as, in their own way, forms of storytelling. They tell stories about what those who produce them and what those who rely on them care about most. These cares are coded into the naming of some variables as important and the exclusion of others, as Bowker and Star (1999) have shown. They are coded into the statistical forms of reason that are thought to produce facts about which interventions worked and which interventions failed, the causes of death, cost effectiveness, and on and on. This is more than saying metrics are politically coded or that numbers can decide fates (of individuals or whole populations) as, at one time, only sovereigns could. Specific numbers can certainly move policy, confer political allegiance, guarantee funding, even bring about health. But they do so not simply by claiming truth about the empirical world. They do so because of the ways they are “produced” and the ways they are circulated. These productivities and circulations are the stories that precede and exceed numerical forms of truth-telling. At the same time, metrical forms of reason, and truth-telling often displace other kinds of knowledge, other forms of evidence (Sangaramoorthy and Benton 2012). Metzl (2010) reminds us that metrical forms of reason and accountability can both conceal and produce empirical facts that work against health. We further this project by studying how producing metrical forms of accountability can displace other activities, other ways of knowing, and other ways of seeing, seeing other things. These are the other kinds of stories that metrics tell.

Similarly, while it is commonplace to fall back on the truism that numbers are inherently *not neutral* in that they can always *be used* for any political purpose, as many authors have shown, it is another thing to show how the act of data collection becomes always and invariably political as a form of knowledge *because* it claims political neutrality. In fact, it takes a lot of work to make something seem politically “neutral,” especially when this as-

sumption undergirds efforts to actually *make* political claims or to set political policy. When the WHO uses metrics to push back on health-jeopardizing political policies (advancing a pro-health politics of their own), they do so by presuming the truthfulness of numbers in and of themselves (Chorev 2012), and this takes a lot of work and a lot of management of evidence in such a way that the numbers do this political work. How metrics, as forms of evidence-making, constitute a politics in their own right by claiming neutrality still needs to be explored ethnographically. Anthropologists, long concerned with the problems of veridication in and around different kinds of evidence (Leibow et al. 2013), have returned to this exploration today by looking at quantification and its epistemic demands on reason and on practice. Our book extends this project by exploring how the production of metrics works in global health today as a political imperative tied to new forms of evidence, old statistical forms of reason, painfully lingering financial arrangements of neoliberalism, and disruptions of notions of national sovereignty that have become visible in our time.

I do not mean by this that numbers can never be helpful to the project of health. On the contrary, the assumption is usually that numbers and the metrics they sustain are needed for improved health and health care policy. But, as scholars like Jean Lave (1988) and Helen Verran (2001) have shown, even the basic notion of counting (long before it is rendered “scientific”) needs to be seen as a learned, culturally specific possibility. People not only count in different ways; they also take counting to mean different things. Why, then, would we assume that the turn to using statistical forms of reasoning since the nineteenth century in order to make sense of social life has been anything but similarly conditional on certain very specific, colonial, postcolonial, and sometimes very neocolonial kinds of arrangements, as Hacking (1990) notes? Similarly we might interrogate how the metrics being pushed today (in relation to statistics, RCTs, and evidence-based medicine) are also conditional on specific political and economic arrangements that hope to transcend old politics and political forms.

As the use of quantitative forms of reasoning has grown in the health and clinical sciences since the nineteenth century, so too has it become harder and harder to imagine that these forms of managing the empirical world are simply artifacts of culture or politics. But they are. Their spread and their refinement during the second half of the so-called postcolonial century and their growth and acceptance as useful to the work of global health in the first decades of the new millennium have gone unchallenged

in too many global health circles. They are not entirely unchallenged, and many of those who work in and through these forms of reason (including economists and statisticians) have debated the utility of one kind of accounting over another, as we will see. Still the rolling out of expectations around statistical and experimental forms of data and evidence has been consistently growing in ways that deserve more scrutiny.

Ethnography

Anthropologists have been saying a lot about these trends in general, and medical anthropology has much to add to this conversation in global health in particular. The chapters here document the specific ways practices of *inventory* and *intervention* through metrics work in detail in global health. As RCTs and statistically robust forms of health work become the gold standard for not only showing effectiveness but also for obtaining funding, so too do these techniques become pathways for other things: for getting elected to political office, for asserting native sovereignty, for making profits.

At the same time, these chapters suggest that the metrics, even while being produced, often produce collateral effects and opportunities that mess up our easy confidence in the numbers. In fact they show that ethnographic research—because it enables a focus on single cases in all of their singular and idiosyncratic complexity but also because it enables a focus on other kinds of evidence writ large—produces empirical data that form not only an evidence base but an evidence base that is sometimes more truthful, proposing alternatives to the evidence-making from metrics work. Ethnographic materials are a potential source of alternative evidence that not only contrasts with the kinds of evidence required for good metrics work but also sometimes unseats its hold on truth.

The metrics of concern in these chapters appear in many ethnographic forms. Sometimes they refer to numerical accountabilities over how many women died or lived after giving birth, and sometimes they refer to more complex modes of numerical figuring: RCTs, power calculations, other kinds of statistics, other kinds of counting. Always the need for numerical data appears as a kind of burden of proof, as if proof lay intrinsically in what the numbers tell us. Each chapter offers evidence about how metrical forms of reason get tangled up in things far beyond health, while preventing other specificities of health to be concealed. Cumulatively we might

say that these stories tell us something about how the new commitments to metrics both work and don't work in the effort to do global health today.

Finally, what can be made of the stories—the empirical events, experiences, and myriad occurrences and facts—that do not lend themselves to being counted, or at least not counted without an epistemological violence being done to the complicated empirical truths they offer? What are we to do with empirical truths about particularities and interwoven strands of cause and effect, of rationality, that seem always beyond the pale of reductive forms of counting—the things that spill over and can't be included in the counting exercises?

For some it is the hallmark of the ethnographic method to try to capture the vast breadth and depth of this complexity of the singular case. The goal is often to capture this complexity without reducing the phenomena observed to simple forms that can be counted in ways that make one case just like another, one specific event comparable to the next. Forms of knowledge that resist reductionism, for many an anthropologist, provide more stable and more reliable forms of truth-telling than those that involve reductionism. But what of the fact that many of us (I would guess), if faced with the opportunity to ensure a single life was saved or to document that a single intervention worked, would still want to “count” this as important enough to change policy, to change the world, even if it carried no statistical or metrical weight of validity? How can we count these complex and singular experiential moments in ways that do not involve reductions in the march toward better metrics in global health? Might these sorts of facts work as a form of affective accountability for metrics exercises, or might they instead work to unseat the claims of the metrics, or might they do both?

The Chapters

Metrics trends in international health have a long history, but it has been only in recent decades, and with the arrival of what is being called *global health*, that a new kind of metrics has been promoted as the sine qua non—an indispensable ingredient—of good global health work. In part I, “Getting Good Numbers,” Claire Wendland and Adeola Oni-Orisan show how the effort to produce numbers for global health statistics is deeply fraught. Both are looking at the case of maternal mortality in African states; both make it very clear that the counting practices used to make claims about

successes and failures in maternal health often place political outcomes over those of health.

Wendland reveals with stunning clarity how statistical accounts of maternal mortality in Malawi enable the production of truths that not only distort empirical reality through estimation practices but also efface massive amounts of information about who is involved with and what really happens during deliveries on the ground. Taking us on an ethnographic tour from the clinic to the world forums where maternal mortality statistics are used to justify state budgets and election campaigns, but also through the statistical forgeries that provide a lexicon for such activities, Wendland reveals the logics that enable conversations about death rates to feel real and solid even while necessarily being deeply flawed. She shows us how important these numbers are and how they become authorized as arbiters of action, even while documenting how hard it is to get “real” numbers that nevertheless must be produced because they serve so many different purposes.

Oni-Orisan reveals similar problems in the production of statistics about maternal mortality in Nigeria. Starting in the clinic and observing a case of maternal death “erasure,” she asks: How does the production of numbers depend on the political accomplishments that those numbers promise? Tracing the effects of the metrics from the invisibilities they leave behind, Oni-Orisan is able to show that numbers are not simply tools used in the political rhetoric of aspiring state bureaucrats; they are enactments of politics in their own right. Political careers are made and destroyed by way of health statistics in this region of Nigeria. Recognized as a global health success by funding agencies that are deeply invested in metrics, health aid becomes a new kind of tool for governance, and political legitimacy trickles down to decision-making and accounting practices at the hospital bedside. These practices, she notes, can mean political success, even when they result in unnecessary mortalities that remain hidden.

Over time and with enough evidence like this, we start to see a picture of omission and erasure — lives that are not saved, programs that have been derailed — that accompany or are enabled by the reliance on metrics. This is a problem not just in one place, with one aberrant program here and there. This is a problem found in many places, in many instances, where numbers are demanded. Exploring the production of numbers up close like this enables one to begin to see their solidity dissolve. Stories about this produce a form of counterevidence and this is a good place to begin this volume.

Part II, “Metrics Politics,” tackles thorny questions about how metrics become integral to practices of governance in the postcolonial era. That is, whereas part I affirmed the ways numbers and their production are political acts, part II explores how metrics become tangled up in state governance and problems of sovereignty. Marlee Tichenor offers a stunning example of this in her study of the data-retention strike in Senegal, in which health workers made citizenship demands on the state (for better pay, working conditions, etc.) by withholding data that the state needed to make its claims about health and budgets for aid. Although we are shown in this chapter not only how “wobbly” the numbers are (as Wendland calls them) because of the conditions of their production, Tichenor also shows us how solid they must be made to seem because of their foundational role in governance in states like Senegal, where large sectors of the economy are tied to the health sector.

In a similar vein Molly Hales shows us how metrics activity becomes entangled in problematic ways with efforts to assert sovereignty on the part of Native Alaskan Yup’ik. She points to the messiness that arises from efforts to document efficacy using metrics, particularly in the case of Native American sovereignty that for at least a few decades has pushed for use of tribal concepts of ethics, health, and community as a basis for designing health interventions and treatments. As Native Alaskan Yup’ik sovereignty shifted in the 1970s toward architectures of neoliberalism, the question of how tribal sovereignty is enacted in this region is doubly vexed by being beholden to fiscal priorities (the tribe must be run at least in part like a business) but also to the problems of funding public sector activities that cannot pay for themselves. In Hales’s pithy explication of these dynamics, we learn of tensions that bubble up over how to manage the production of evidence when the stakes of using some kinds of evidence over others are deeply tied to the very vision of sovereignty (and the fiscal tactics used for this) that locals care the most about.

In part III, “Metrics Economics,” Susan Erikson and Lily Walkover offer compelling evidence that the burdens of a new emphasis on metrics are meted out in and through complex orchestrations of finance, producing accountabilities that are as much about fiscal profit and survival as they are about health. Erikson begins with a scathing exploration of the linkages between creative financing and profit-driven solutions that are being pushed by the big players in global health funding today, teasing out the threads that have sutured together corporate interests and investment models to

the humanitarian work of global health. Walking in step with my own exploration in chapter one, Erikson takes us down into the nitty-gritty details of these neoliberal relationships. New partnerships between public and private entities, investment portfolios and humanitarian missions, profit-making and health outcomes are all enabled by the use of metrics, and the particular fiscal architectures that are being celebrated up the line by global health planners and financiers become highly questionable in a number of ways. When funds for health programs are derived by way of investments in companies that are also responsible for disease, we might question whether this arrangement is truly a win-win arrangement. Perhaps unraveling the role of metrics in all this will help us to unravel, or at least make visible, what is at stake in some of the more questionable practices in this set of arrangements.

In affirmation of the cartography of market-driven global health mapped out by Erikson, Walkover offers a troubling example of the outcomes of metrics-driven demands at a nonprofit. She also suggests an opportunity to think of how to change the discourse. Hesperian Health Guides, which has arguably made one of the largest impacts in global health over the entire era of postcolonial health development through its *Where There Is No Doctor* books, confronts the new demands for metrics, the same metrics Erikson points to in her study of finance in global health. Walkover shows how these demands put organizations like Hesperian at risk. By tracing the organizations' ethos and practices that have traditionally been grassroots and bottom-up in ways that do not easily lend themselves to being statistically countable, Walkover maps out exactly what is at stake in trying to meet the demand for new metrics data. In searching for better metrics as evidence of their success, the very practices that have made these organizations successful are often derailed. Remaining accountable to communities is often undermined by requests to make communities "countable" for the benefit of potential funders.

Part IV, "Storied Metrics," thus explores the proposition that is implicit in the chapters up to this point: that we might read the metrics as storied in their own ways. However, the chapters here tell us a good deal about how people working within metrics systems work against the idea that metrics tell only one kind of story. Carolyn Smith-Morris takes up the issue of "fidelity" that is used in metrics work as a form of veridication (establishing that the data are reliable because they were collected with fidelity to the protocol). Looking at how ethnographers worked with RCT research

on veterans with spinal cord injuries in the U.S. Veterans Hospital Administration, Smith-Morris shows that achieving fidelity does not necessarily mean that RCT work is capturing the kind of evidence that actually reveals why the program works or does not work. Efforts often fail to capture these messy and uncountable details that really make a difference in veterans' lives. Still, she shows, statistical accounts are used to move policy forward and to justify one kind of program over others. Her work suggests that statistical practices might actually be read as social processes and entanglements that enable these programs to claim success and also constitute their existential power on the ground, in the clinic. How researchers like Smith-Morris, who are on the front lines of RCT anthropological engagements in health care, are actually interrogating and changing the work of RCT metrics is a topic we have much to learn from in relation to critique and also in relation to productive opportunities. This work, she implies, forms a different lexicon for speaking not only about outcomes but also about fidelity in research.

Another way to think about the storied metrics of global health is through what Pierre Minn offers; he shows that as global health interventions are increasingly expressed in terms of quantitative outcomes and “deliverables,” individuals and organizations whose activities fall outside the purview of these units must find ways to communicate their actions to funders and supporters, a theme also touched on in Hales's and Walkover's chapters. Studying Konbit Sante, a U.S.-based NGO working in northern Haiti whose mandate explicitly eschews clinical interventions and the development of a “parallel health system,” Minn illustrates how staff and volunteers draw on the language of alterity and distinction to gain visibility and support for their programs. The organization's emphasis on the relational aspects of their work has proven to be a double-edged sword, both limiting opportunities for expansion while earning them a prominent reputation among actors in Haiti's health sector. As they work to represent their accomplishments in the quantified terms that meet the demands of transnational aid bureaucracies (“widgets”) in ways that resonate with the affective sensibilities of individual North American donors, they also accomplish a kind of global health that speaks back to power in and through a critical engagement with metrics.

As closure for the volume, I offer a brief epilogue that takes up the question of what counts as good global health work under the pressures of global health metrics today. Instead of reiterating the flaws of metrical forms of

accountability, I discuss a few novel propositions being put forth by colleagues and practitioners who are in their own ways toying with the problem of numbers and answering those who question the tendency toward tyranny in the global health conversations we are having. Overall my hope is that readers will be called to consider how else we might think about the architectures of global health in terms of who benefits and who does not, about what we mean by evidence in the steady march toward better forms of accountability, and to embrace the possibility of plural forms of deliberation over what we mean by both evidence and efficacy at all in global health.